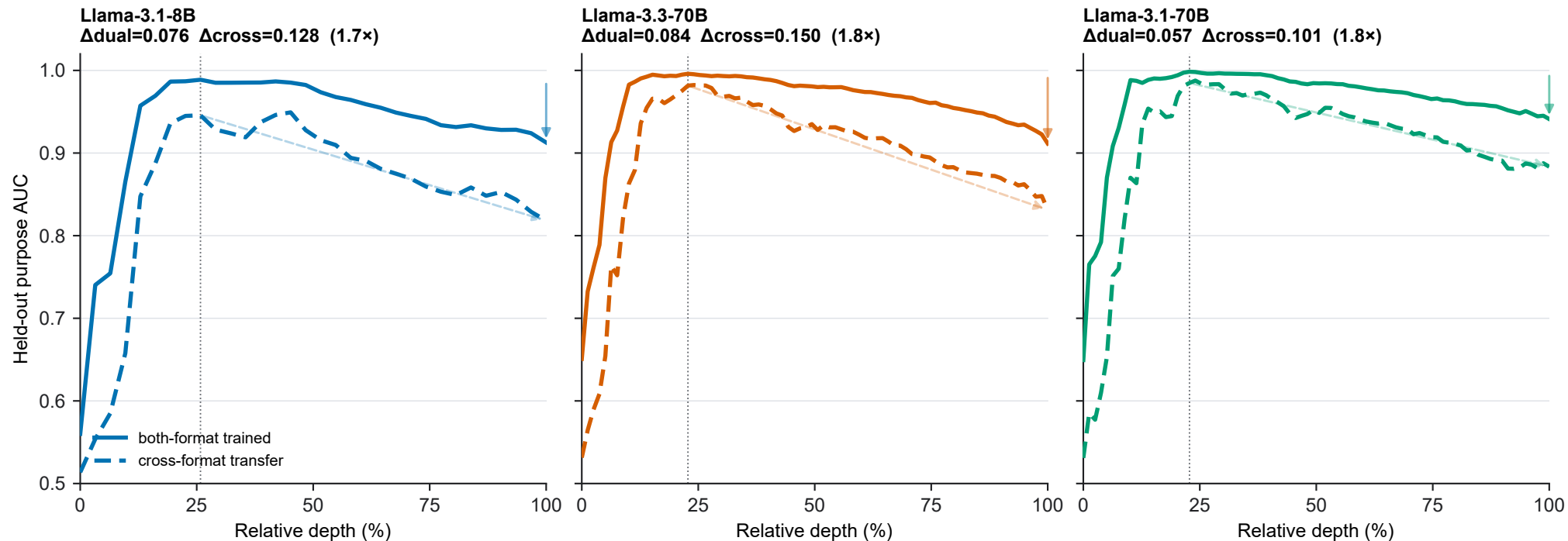


Cross-format-transferred probes lose disproportionately more AUC in late layers



From each model's decorrelated-AUC peak (dotted vertical line) to the final layer: the cross-format-transferred probe (dashed) loses ~1.7–1.8 $\times$ more AUC than the jointly-trained probe (solid), consistently in all 3 models. Reading: late transformer layers reorganize toward output/next-token computation; a probe trained on both formats can track that reorganization even as it becomes more format-conditional, but a probe trained on one format alone cannot follow a shift it never saw — so transfer degrades faster than joint decodability, and the gap between them widens. Descriptive, not causal; unlike the two geometry checks in this project, not a correlation between depth-trending curves.